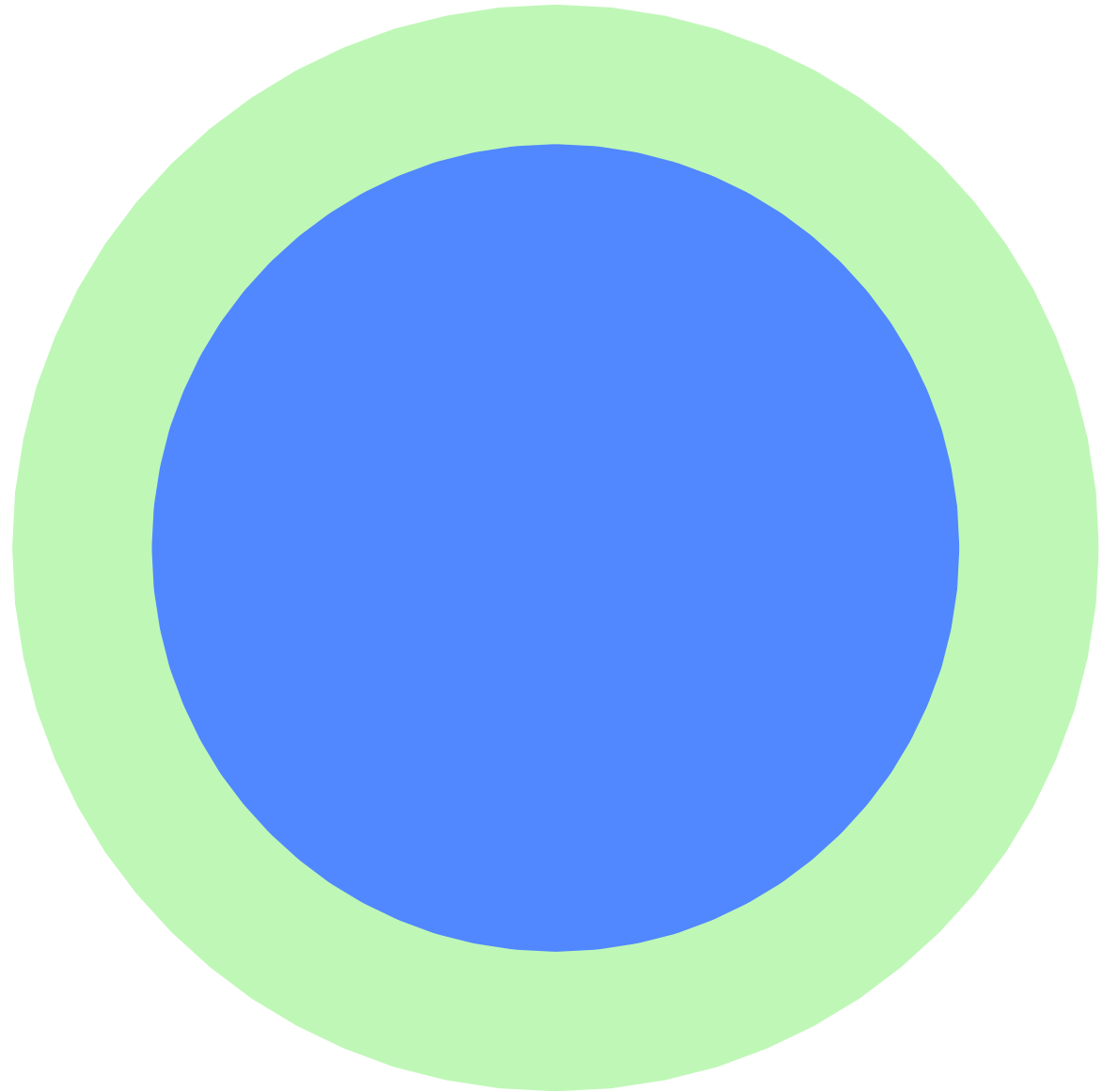

SyncraSong

*A cross-platform tool to
transfer music playlists.*



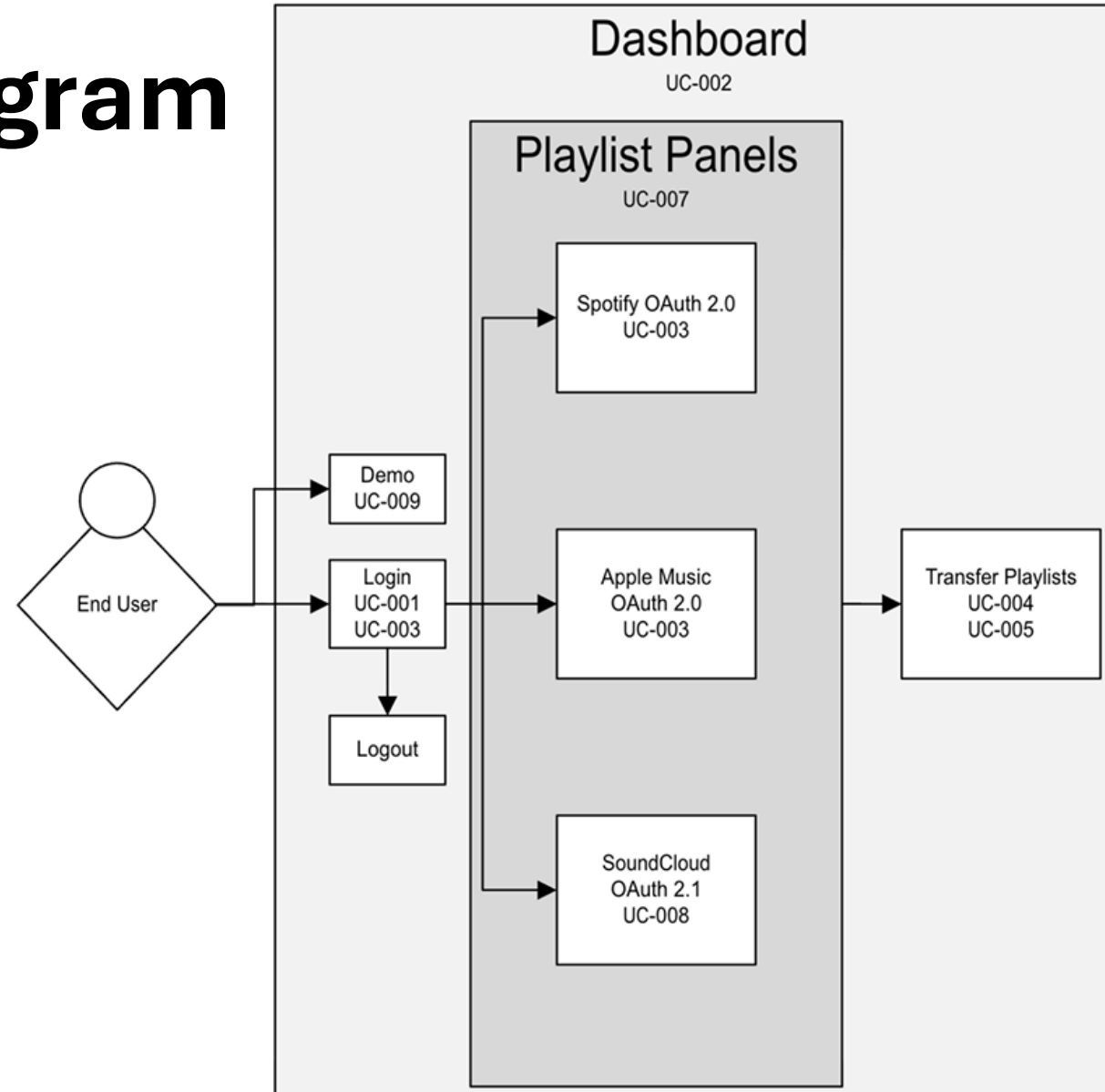
Problem Statement & Motivations

- Music listeners want to change their music platform.
- Services do exist but they are paid services.
- Goal: Seamless migration while minimizing loss of songs.

Core Features

- User Login: OAuth 2.0
- Accessibility: Web access
- Transfer playlists between selected platforms
- Load Balancing: Queue & Job process in the backend

User Diagram



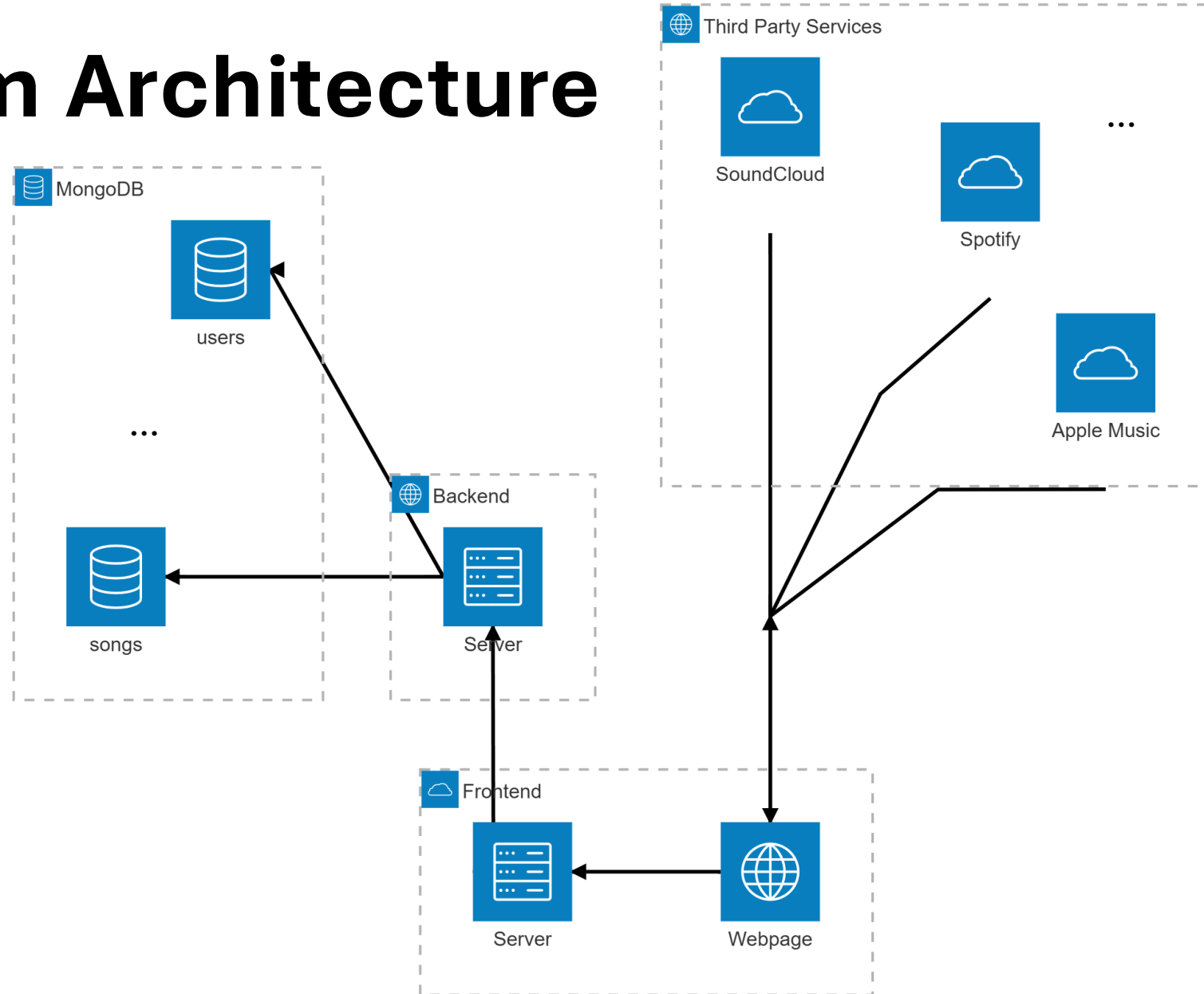
Backlog

- As a user,
 - I want to login using Spotify or Apple Music.
 - I expect my playlists to transfer between services.
- As a new visitor,
 - I want a guidance on how to use the platform.
 - I don't want to create a new account on your website.
- As the site manager,
 - I don't want my tools to be reliable and available majority of the time.

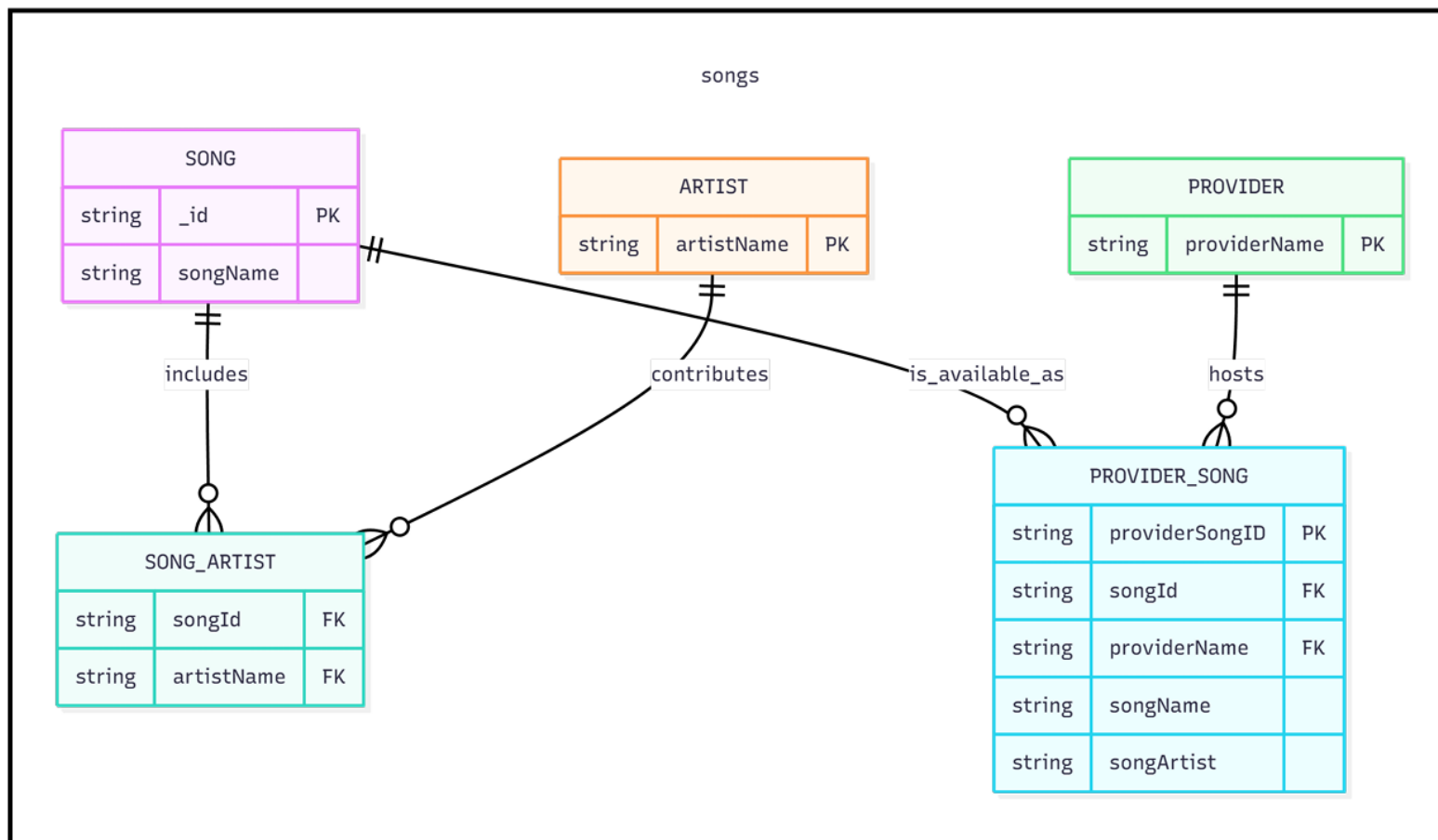
Architecture

- As a user,
 - I want to login using Spotify or Apple Music.
 - I expect my playlists to transfer between services.
- As a new visitor,
 - I want a guidance on how to use the platform.
 - I don't want to create a new account on your website.
- As the site manager,
 - I don't want my tools to be reliable and available majority of the time.

System Architecture



Database: Relational Caching



Example Entry:

```
_id: ObjectId('6857c49bf2c7c88876aa3b94')
songName: "clarity"
▼ songArtists: Array (2)
  0: "zedd"
  1: "foxes"
▼ providers: Array (2)
  0: "spotify"
  1: "apple"
▼ spotify: Object
  songID: "60wwxj6Dd9NJlirf84wr2c"
  songName: "Clarity"
  songArtist: "Zedd, Foxes"
▼ apple: Object
  songID: "1440818495"
  songName: "Clarity (feat. Foxes)"
  songArtist: "Zedd"
```




SyncraSong



Apple Music



Last updated:
11/9/2025, 4:01:19 PM [REFRESH](#)



Blanke: The Complete...
110 songs



coast line (SyncraSong)
150 songs



56 songs



cupid arrows
116 songs

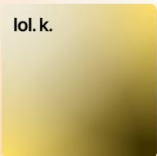


No Image

discump
1644 songs



Favorite Songs
1 songs



lol. k.
573 songs



loud noises
176 songs



melatones
46 songs



mini headbangers
96 songs

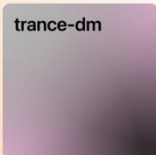


No Image

nostalgia
37 songs



power outage
2119 songs

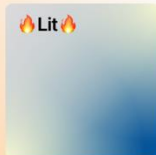


trance-dm
11 songs



No Image

vibe
37 songs



Lit
296 songs



7



Spotify



Last updated:
11/9/2025, 4:01:19 PM [REFRESH](#)



trance-dm
11 songs



Blanke: The Complete...
114 songs



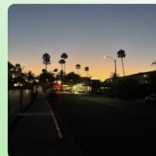
cupid arrows
119 songs



loud noises
208 songs



melatones
46 songs



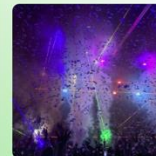
iv summer
34 songs



sf nostalgia drive
43 songs



7
178 songs



mini headbangers
98 songs



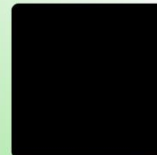
vibez
200 songs



lol. k.
644 songs



void
244 songs



transpire
85 songs



coast line
150 songs



crossfaded :)
314 songs



SyncraSong



Apple Music



Last updated:
11/9/2025, 4:01:19 PM [REFRESH](#)



Blanke: The Complete...
110 songs



coast line
(SyncraSong)
150 songs



56 songs



cupid arrows
116 songs



No Image
discump
1644 songs



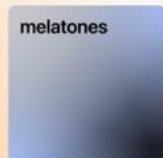
Favorite Songs
1 songs



lol. k.
573 songs



loud noises
176 songs



melatones
46 songs



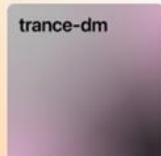
mini headbangers
96 songs



No Image
nostalgia
37 songs



power outage
2119 songs



trance-dm
11 songs



No Image
vibe
37 songs



Lit
296 songs

7



Spotify



Last updated:
11/9/2025, 4:01:19 PM [REFRESH](#)

Pending Playlists

TRANSFERRING...



56 songs

Commit Transfer

Cancel All

First Prototype Results:

- Uncached playlist (n = 100 songs): ~ 41 seconds.
- Cached playlist (n = 100 songs): ~6 seconds.
- Spotify → Apple Music: On average 29 seconds.
- Apple Music → Spotify: On average 53 seconds.

Overall, I was pleased with these results.

Problems:

- Apple Music API required 2 calls for accurate results
 - Increases transfer time by 155 ms per song (i.e., 15.5 seconds for 100 songs)
- Adding services in the backend was not scalable as it was first hard coded in JavaScript.
- Working indicators if songs are transferring or have failed.
- Website is not responsive. Not phone compatible.
- UI is ugly, I'm not great in UI design.

Addressing the Problems

- Hired UI/UX designer to redesign my website.
- Rewrote the backend using TypeScript, scraped JS repository.
- Added jobs and queue system.

SyncraSong

Apple Music



Spotify



Drag playlists from either platform to the transfer area

Confirm Transfer

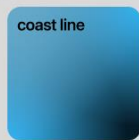
Cancel All

Apple Music

Select All



Blanket: The Complete Discography
110 songs



coast line
151 songs



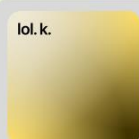
56 songs



No Image
discump
1644 songs



Favorite Songs
1 songs



lol. k.
573 songs



loud noises
176 songs



melatones
46 songs



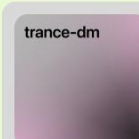
mini headbangers
96 songs



No Image
nostalgia
37 songs



power outage
2119 songs



trance-dm
11 songs

Spotify

Select All



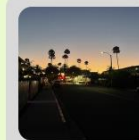
Blanket: The Comple...
114 songs



cupid arrows
119 songs



melatones
46 songs



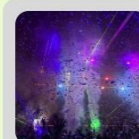
iv summer
34 songs



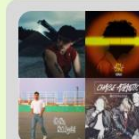
sf nostalgia drive
43 songs



7
178 songs



mini headbangers
99 songs



vibez
200 songs



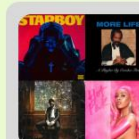
lol. k.
645 songs



void
244 songs



coast line
151 songs



crossfaded :)
314 songs

```
_id: ObjectId('6920ddca84acdaef2571ae28')
userId: "austinkim"
▼ source: Object
  provider: "soundcloud"
  playlistId: "2117949857"
▼ target: Object
  provider: "spotify"
▼ options: Object
  status: "success"
  attempts: 0
  runAt: 2025-11-21T21:46:50.932+00:00
  createdAt: 2025-11-21T21:46:50.932+00:00
  updatedAt: 2025-11-21T21:47:57.013+00:00
▼ meta: Object
  phase: "complete"
  ▼ source: Object
    provider: "soundcloud"
    playlistId: "2117949857"
    playlistName: "coast line"
    totalTracks: 145
  ▼ matching: Object
    requested: 134
    matched: 134
    unmatched: 0
    missingIsrc: 11
  ▶ missingIsrc: Array (11)
  ▶ unmatched: Array (empty)
  ▶ target: Object
    added: 134
  ▼ summary: Object
    totalTracks: 145
    transferred: 134
    skippedMissingIsrc: 11
    unmatched: 0
    destinationPlaylistId: "6gTHrMzz1j0OvwfVFYStVu"
totalTracks: 145
transferredTracks: 134
```

Next Steps & Plans:

- Implement more platforms.
- Optimize algorithms or look for alternative ways to fetch.
- Load balancing test: How many concurrent playlists can it transfer?
- User preference saving (e.g., dark mode).
- User unlinking platforms.
- and more...

Demo:

